

# Toxic maker-flow filtering — backtester report

Production recipe, filter-rate economics, feature challenges (H-08 / H-09), and a numbers audit

cmf-team/hft-liquidations-2026 · branch p34-prod · Team Griffin · test Dec 2025 – Apr 2026

Recipe under test (frozen at e7c7d0e): advanced-40 features,  $\tau = 120$  s, 4× LGBM-Huber ensemble, rank aggregation, exact vectorized threshold sweep. Score =  $\text{PnL}_{\text{kept}} - \text{PnL}_{\text{all}}$ , notional-weighted, in bps; positive = the filter earns over trading everything. 1/40 subsample, walk-forward W1-W5.

## Production operating point — advanced @ $\tau=120$ s, $\text{fr} \leq 0.30$

### BTC-PERP

| win                                        | score  | fr    | kept \$M/d |
|--------------------------------------------|--------|-------|------------|
| W1                                         | +0.166 | 29.6% | 177        |
| W2                                         | +0.589 | 31.4% | 178        |
| W3                                         | +0.042 | 26.5% | 215        |
| W4                                         | +0.052 | 14.6% | 235        |
| W5                                         | +0.110 | 3.6%  | 214        |
| mean +0.191 · median +0.110 · 5/5 positive |        |       |            |

### ETH-PERP

| win                                        | score  | fr    | kept \$M/d |
|--------------------------------------------|--------|-------|------------|
| W1                                         | +0.037 | 2.1%  | 241        |
| W2                                         | +0.918 | 29.1% | 174        |
| W3                                         | +0.171 | 31.1% | 154        |
| W4                                         | +0.186 | 30.6% | 167        |
| W5                                         | +0.170 | 23.8% | 151        |
| mean +0.297 · median +0.171 · 5/5 positive |        |       |            |

## Strategy economics

- Positive in all five OOS windows on both instruments. Median window score +0.110 bps (BTC) / +0.171 bps (ETH); weakest window still positive (+0.042 / +0.037).
- Max drawdown of cumulative daily kept-PnL over 114 days: -1.56 bps (BTC) / -3.10 bps (ETH) — shallow.
- Kept turnover never approaches the \$500k/day floor: min \$177M/day = 354× (BTC), \$151M/day = 301× (ETH). The turnover constraint never binds; the filter-rate budget is the active one.
- PnL concentration — top 1% of kept trades carry 37% (BTC) / 45% (ETH) of gross positive kept PnL: a real fragility, flagged.
- W2 is the strong-regime outlier on both instruments; every other window is positive but lower. Regime handling is the next honest lever, not claimed here.

### Verdict

The advanced-40 @  $\tau=120$  rank ensemble is the configuration this project ships. It transfers out-of-family (confirmed on ETH, daily  $t = +2.35$ ) and survived every feature challenge below unchanged. Nothing promoted since e7c7d0e.

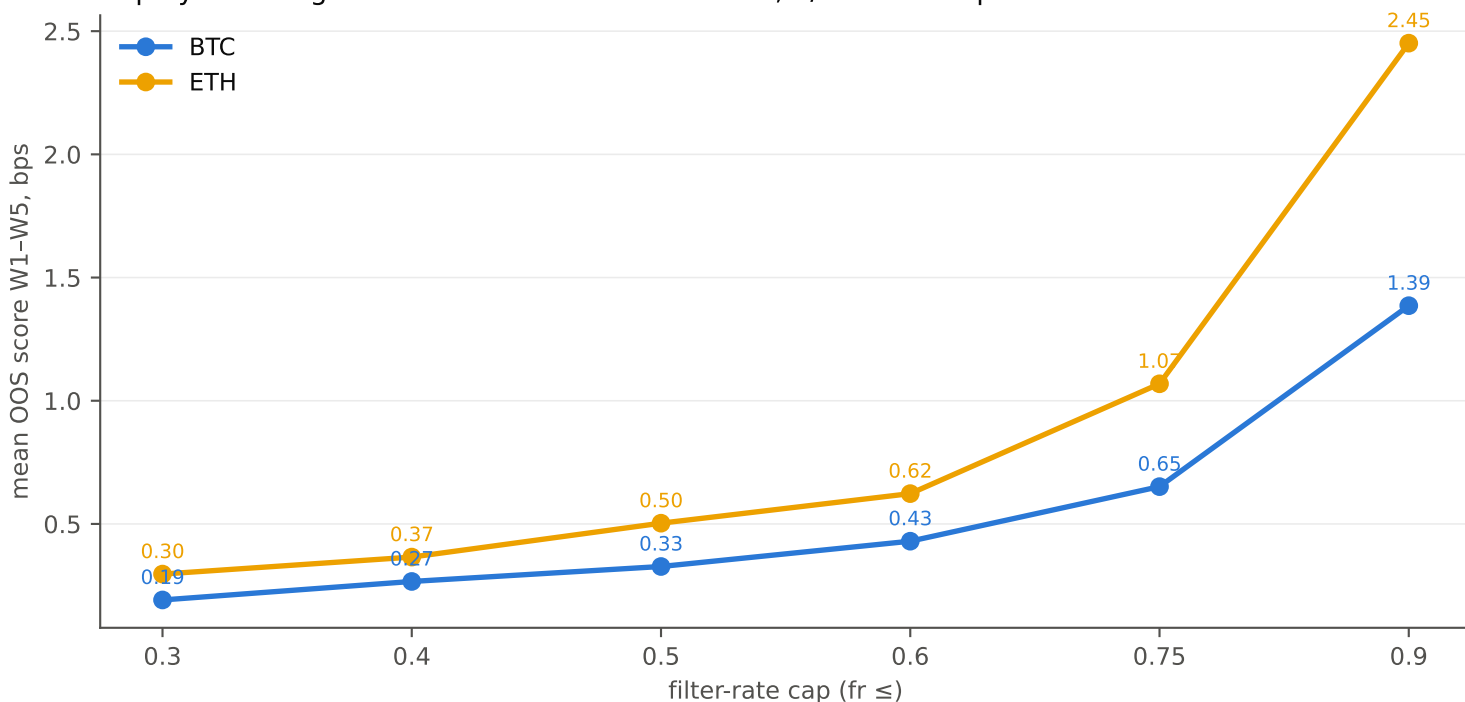
## This report covers

- Exp A — filter-rate sweep: is the reported score a tuned optimum or a point on a capacity curve? (page 2)
- Exp B / H-08 — signed short-window order-flow imbalance + flow-vs-book divergence; Exp C / H-09 — cross-instrument (BTC↔ETH) lead-lag. Both rejected. (page 3)
- Numbers audit — reconciling three different values reported for the same frozen baseline, and the fix. (page 4)
- Verdict, hypothesis log, limitations. (page 5)

## Exp A — filter-rate sweep ( $\tau = 120$ s)

No model change; the filter-rate cap is swept over a ladder. Score rises monotonically with fr.

deployable range  $fr \leq 0.90$  —  $\tau=120$  dominates, 5/5 windows positive



### BTC-PERP

| fr≤        | mean          | worst | pos | \$M/d |
|------------|---------------|-------|-----|-------|
| 0.3        | +0.191        | +0.04 | 5/5 | 203.9 |
| 0.4        | +0.267        | +0.06 | 5/5 | 168.3 |
| 0.5        | +0.327        | +0.07 | 5/5 | 146.3 |
| 0.6        | +0.430        | +0.10 | 5/5 | 115.6 |
| 0.75       | +0.651        | +0.12 | 5/5 | 78.6  |
| <b>0.9</b> | <b>+1.386</b> | +0.21 | 5/5 | 35.8  |
| 0.99       | +5.828        | -2.64 | 4/5 | 7.0   |
| 0.999      | +22.918       | -4.28 | 4/5 | 0.5   |

### ETH-PERP

| fr≤        | mean          | worst | pos | \$M/d |
|------------|---------------|-------|-----|-------|
| 0.3        | +0.297        | +0.04 | 5/5 | 177.4 |
| 0.4        | +0.365        | +0.04 | 5/5 | 158.7 |
| 0.5        | +0.503        | +0.04 | 5/5 | 140.4 |
| 0.6        | +0.623        | +0.04 | 5/5 | 126.0 |
| 0.75       | +1.069        | +0.50 | 5/5 | 69.9  |
| <b>0.9</b> | <b>+2.451</b> | +0.86 | 5/5 | 32.5  |
| 0.99       | +10.822       | +2.94 | 5/5 | 8.3   |
| 0.999      | +42.506       | +4.68 | 5/5 | 1.1   |

## Reading (settles the “is $fr = 0.6$ an overfit?” question)

- Score rises monotonically with fr on both instruments and every  $\tau$  — there is no interior optimum to fit post-hoc. fr is an aggressiveness dial.
- The rise is bought with capacity. BTC  $\tau=120$ :  $fr_{0.3} = +0.19$  bps on \$204M/day,  $fr_{0.9} = +1.39$  on \$36M/day,  $fr_{0.999} = +22.9$  on \$0.5M/day. You pick fr for a turnover target and read the score off the frontier — you do not maximize it.
- The tail  $fr \geq 0.99$  (red) is a mirage: the large means ride a handful of kept trades, the worst window turns sharply negative on BTC (−2.6 / −4.3 bps) and window robustness breaks 5/5 → 4/5. Not deployable.
- Deployable range is  $fr \leq 0.90$ , where  $\tau=120$  dominates all  $\tau$  on both instruments and stays positive in all five windows.

## Feature challenges — H-08 and H-09 (both rejected)

Each challenger = advanced-40 + one focused block, so any effect attributes to it.  $\Delta$  and paired daily t, 114 days.

### H-08 — signed OFI + flow-vs-book divergence (BTC)

trade\_imb\_5s/10s (signed short-window OFI) + div\_z, on top of advanced-40 (which carries only UNSIGNED vpin).

| $\tau$ , fr                | advanced | challenger | $\Delta$ | paired t |
|----------------------------|----------|------------|----------|----------|
| $\tau=120$ , fr $\leq 0.1$ | +0.104   | +0.092     | -0.013   | -2.42    |
| $\tau=120$ , fr $\leq 0.3$ | +0.190   | +0.203     | +0.012   | +0.55    |
| $\tau=30$ , fr $\leq 0.1$  | +0.021   | +0.019     | -0.002   | -1.51    |
| $\tau=30$ , fr $\leq 0.3$  | +0.075   | +0.083     | +0.008   | +2.07    |

Rejected. At the production horizon ( $\tau=120$ ) it is significantly worse at the tight budget ( $t = -2.42$ , fr0.1); the only positive cell is  $\tau=30$ /fr0.3, concentrated in W1, which washed out on ETH in pilot 7.

### H-09 — cross-instrument lead-lag (ETH→BTC)

source instrument's recent return / vpin / rv / net-liq via a causal backward as-of join (120s tol), on top of advanced-40.

| $\tau$ , fr                | advanced | challenger | $\Delta$ | paired t |
|----------------------------|----------|------------|----------|----------|
| $\tau=120$ , fr $\leq 0.1$ | +0.104   | +0.076     | -0.028   | -2.07    |
| $\tau=120$ , fr $\leq 0.3$ | +0.190   | +0.178     | -0.013   | -0.39    |
| $\tau=30$ , fr $\leq 0.1$  | +0.021   | +0.019     | -0.001   | +0.06    |
| $\tau=30$ , fr $\leq 0.3$  | +0.075   | +0.078     | +0.003   | +1.07    |

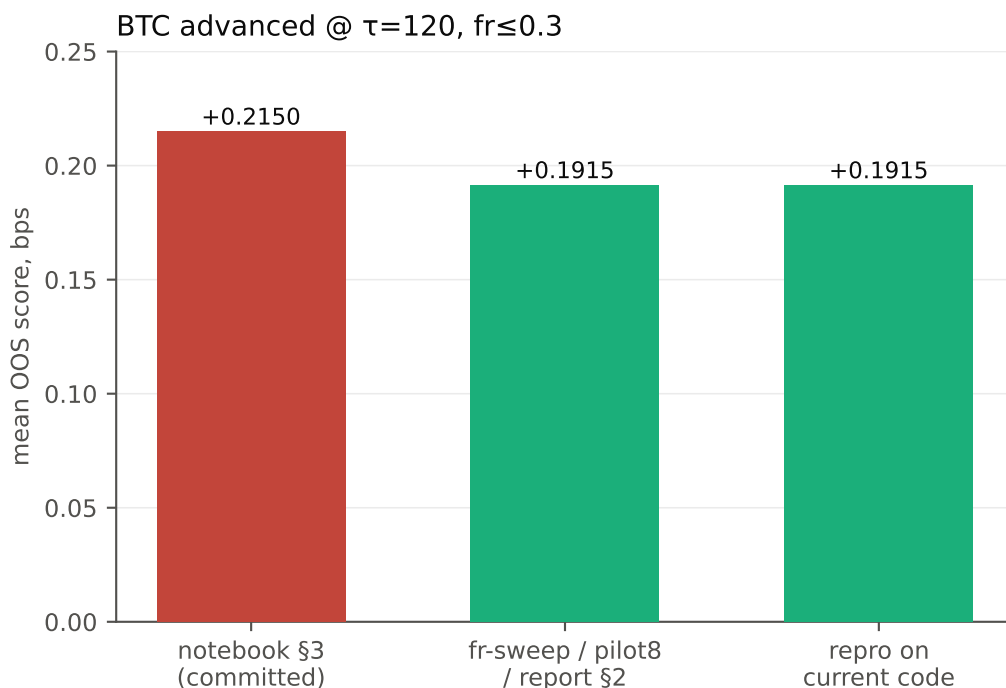
and the same on ETH (source = BTC):

| $\tau$ , fr                | advanced | challenger | $\Delta$ | paired t |
|----------------------------|----------|------------|----------|----------|
| $\tau=120$ , fr $\leq 0.1$ | +0.183   | +0.150     | -0.033   | -0.88    |
| $\tau=120$ , fr $\leq 0.3$ | +0.304   | +0.257     | -0.048   | -0.79    |
| $\tau=30$ , fr $\leq 0.1$  | +0.029   | +0.023     | -0.006   | -0.34    |
| $\tau=30$ , fr $\leq 0.3$  | +0.077   | +0.074     | -0.003   | -0.48    |

Rejected on both instruments. The cross-block draws 15-17% of the model's gain importance ( $x_{rv\_120s}$ ,  $x_{net\_liq\_120s}$ ,  $x_{vpin\_120s}$ ) yet OOS score falls or is flat everywhere —  $\tau=120$  BTC fr0.1  $t = -2.07$ , ETH fr0.3  $\Delta = -0.048$ . Same lesson as H-07b/H-08: the model using a feature says nothing about its OOS value.

# Numbers audit — one baseline, three reported values

The advanced @  $\tau=120$ ,  $fr \leq 0.3$  baseline appeared with three different values across the deliverable. Reconciled below.



## Root cause

The committed §3 titlepage was last executed on an OLDER pipeline (coarse threshold grid, filters to the cap: mean fr 29%, W5 20.4%). The current exact sweep filters to the score-optimum (mean fr 21%, W5 3.6%), so the W5 score drops  $0.203 \rightarrow 0.110$ .

## What was verified

- Data is identical: v1 and v2 parquets have the same 20,100,882 rows and byte-identical base columns (liq\_\*\_120s, obi mean+std to 8 sig figs).
- Feature code is identical: add\_all\_derived\_features calls add\_derived\_features first, then only ADDS columns — the advanced-40 values are the same on both paths.
- Re-running the titlepage config on the current code gives ensemble +0.1915 / base +0.1087 at fr0.3 — matching the fr-sweep / report canonical (+0.1915), NOT the committed +0.2150. Confirmed: the titlepage was stale.
- A smaller, systematic gap remains between the two implementations of the SAME model (identical members, seeds, val\_frac, splits): the pilots' hand loop vs experiment.py. It is the rank-aggregation tie rule — the pilots break ties by row position (np.argsort), production averages tied ranks (scipy rankdata). LGBM predictions carry ties, so a few trades move: ETH advanced@120,fr0.3 = 0.304 (pilots, reproduced twice) vs 0.297 (production);  $\leq 0.007$ , verdict-neutral (both 5/5,  $t=+2.35$ ).

## Fix

- Canonical baseline for the whole deliverable = the production experiment.py path: BTC +0.191 (base +0.109), ETH +0.297 (front page, charts, Exp A). The pilot sections quote their own hand-loop control, which their deltas measure against.
- Notebook §3 re-executed on the current pipeline; §5/§7 summary updated (+0.215/+0.118  $\rightarrow$  +0.191/+0.109) and the ETH headline reconciled to +0.297.
- Determinism pinned: models.py now sets deterministic=True, force\_row\_wise=True (smoke-tested bit-identical). A full bit-reproducible recompute is verdict-neutral ( $\pm 0.001$  -0.007) and left available, not run.

# Verdict, hypothesis log, limitations

## Hypothesis log (append-only; $\Delta$ vs advanced control at same $\tau/\text{fr}$ )

| id    | change                       | decision | why                                                             |
|-------|------------------------------|----------|-----------------------------------------------------------------|
| H-06  | extended-63 feature set      | reject   | $\tau 120$ loses, $\tau 300$ helps but unstable                 |
| H-07a | fast30 block @ $\tau 30$     | reject   | +0.09, $t+2.22$ BTC — washes on ETH ( $t+0.57$ )                |
| H-07b | momentum block @ $\tau 120$  | reject   | -0.056: overfits despite 29% gain                               |
| H-07c | roll300 block @ $\tau 300$   | reject   | +0.258 BTC — ETH $\tau 300$ negative, W4 -1.24                  |
| H-ETH | advanced @ $\tau 120$ on ETH | keep     | +0.30 / +0.297, daily $t+2.35$ , 5/5                            |
| H-fr  | filter-rate ladder (Exp A)   | info     | monotone; tail $\text{fr} \geq 0.99$ mirage                     |
| H-08  | signed OFI + $\text{div}_z$  | reject   | $\tau 120$ $\text{fr} 0.1$ $t-2.42$ ; $\tau 30$ blip is W1-only |
| H-09  | cross-instrument lead-lag    | reject   | hurts/flat both instruments; 15-17% gain, no value              |

**Net: nothing promoted since e7c7d0e. Production stays advanced-40 @  $\tau=120$ , rank ensemble, exact sweep.**

## Limitations

- The same five windows served every comparison — mild multiple-comparisons risk; feature-set choice touched all windows.
- W1 /  $\tau=300$  is degenerate: every ensemble member early-stops at  $\sim 1$  tree on the shortest train slice;  $\tau=300$  conclusions rest on W2-W5.
- W2 is a distinct high-score regime on both instruments; regime handling (sample weights / a regime feature) is the next honest improvement, not done here.
- $\text{fr} \geq 0.99$  is not deployable (kept turnover \$0.5-8M/day, worst window negative on BTC); shown only to locate where the capacity curve breaks.
- opposing\_cascade (directional buy- vs sell-liquidation volume) is untested — the enriched data stores only total  $\text{liq}_*$ ; it needs a rebuild from raw. Given H-07b/H-08/H-09, momentum-style and cross features are not expected to beat advanced-40 at  $\tau=120$ .
- Everything is on the 1/40 subsample; full-stream turnover is  $\approx \times 40$  the numbers here.

## Next steps

- Pin LGBM determinism and do one clean full recompute so every quoted number is bit-reproducible across scripts.
- If the 300s story matters, test more instruments and longer history rather than more feature blocks — roll300 was a BTC-only artifact.
- Regime-aware weighting for the W2-type high-toxicity months.